

Course	SOCSC-UH 1010Q Statistics for the Social and Behavioral Sciences Credits (4)
Professor	Orsola Torrisi
Term	Fall 2023
Lecture Days	Tuesday-Thursday
Lecture Times	Section 5: 5-6.15 PM
Lecture Locations:	C2 E050
Prerequisites:	MATH-UH 1000A or Math Proficiency Assessment
Cross-listings:	Minors > Economics Pre-Professional Courses > Premedical and Health Studies Core Curriculum > Quantitative Reasoning Majors > Political Science Majors > Social Research and Public Policy Majors > Economics Majors > Business, Organizations and Society Majors > Business, Organizations and Society > Required Social Science Courses

Contact Details:

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Course Description:

This course introduces students to the basics of statistics and probability, which are building blocks for much of the research that is done in Social Science. The course gives students an overview of descriptive and inferential statistics, with emphasis on conceptual understanding and statistical thinking. There is heavy concentration on the big picture: What questions are we trying to address and what can we conclude from our analysis. Throughout the course real data from several social science disciplines will be integrated and studied using state of the art software tools. At the end of the semester students should know the difference between populations and samples, how to design experiments, how to summarize and analyze the results of a statistical experiment, how to test basic hypotheses and how to communicate statistical insights.

Course Learning Outcomes (CLOs):

By the end of the course, students will have command of descriptive and inferential statistics (**CLO1**). Specifically, students will acquire the ability to distinguish between populations and samples, develop skills in experimental design, learn how to summarize and analyze statistical findings, gain proficiency in fundamental hypotheses testing, and effectively communicate statistical insights.

Furthermore, students will have command of statistical program (STATA) and coding (**CLO2**). In particular, students are able to write a do-file and execute it properly.

Students will thus contribute to the development and enhancement of the following program-specific learning outcomes (PLOs):

- Critical Thinking: Students can analyze the essentials of a problem logically and independently. They understand basic micro- and macroeconomic concepts and can relate them to real life situations. They can choose and execute modeling strategies with guidance. (Econ PLO1, high)
 - Proficiency in Empirical analysis: This extends the critical thinking capacity and requires knowledge of current statistical software and data management tools. Students understand economic history and economic theories sufficiently to be able to formulate testable hypotheses. They can collect and use economic data from a wide variety of sources to test the validity of hypothesized relationships empirically with relatively little guidance (Econ PLO6, high)
 - Project Management: Students can work towards solutions with persistence and relatively little guidance. They know how to manage their time and resources effectively and collaborate with team members smoothly (Econ PLO3, high)
 - Capacity to engage the professional literature and acquire familiarity with analytical concepts and principles that are essential for political thinking and understanding of political phenomena (Political Science PLO1, medium)
 - Information technology skills to collect data, present, and document empirical regularities in a wide range of social phenomena, with a particular emphasis on politics (Political Science PLO5, medium)
 - Critical thinking, writing, and analysis related to the social science literature; ability to comprehensively and critically analyze the literature in fields of inquiry important for public policy in writing and for oral presentations, with an emphasis on analytical clarity (SRPP PLO3, medium)
 - Critical evaluation of methodological approaches in the social sciences, including qualitative approaches such as in-depth interviewing and ethnography as well as quantitative approaches such as survey and experimental research; in-depth understanding of weaknesses and strengths of the major approaches to social research; familiarity with related software applications (SRPP PLO6, medium)
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- Abilities and skills necessary to design, plan, and carry out a research project independently; includes developing a research question, formulating hypotheses, and designing the empirical approach, implementing the design and collecting data and testing hypotheses (SRPP PLO8, medium)

Teaching and Learning Methodologies:

The course will be divided into lecture sessions where most concepts will be explained to students using theory, examples, games, etc. During tutorials TA's will initiate students in the use of statistical software.

Course Materials:

We will use the 10th edition of "Introduction to the Practice of Statistics" by David Moore, George McCabe, and Bruce Craig (IPS), WH Freeman: New York, 2021.

Furthermore, we will use Brightspace as an electronic resource for our course. Course notes and presentations will be made available to students.

Additional reference: 15th edition of "Statistics for Business and Economics" by James T. McClave, P. George Benson, and Terry T Sincich, Pearson 2021

Software: Stata, Excel

Assignments:

Students will be graded based on a variety of assignments, exams, and participation.

There will be 4 quizzes during the term on the dates specified in the schedule below. They will test students' progress in assimilating concepts. Furthermore, there will be 3 STATA assignments with a combined weight of one quiz. All quizzes and STATA grade will contribute 50% to the final grade. There will be no make-up quizzes. Quizzes can **only** be dropped due to medical reasons (in this case, please provide a doctor's note that clearly states the motive of student's absence).

Students will also work in teams of 2-4 on a project throughout the course. They will have to research data and elaborate statistical analysis on a topic of their choice. Their findings will be presented in class or during recitation. The project will account for 20% of the final grade. It is essential that you work together with a group and participate in the recitations. The project will ask you to complete computational exercises based on the material being covered in lectures and recitations. Ideally, groups have to be composed of students from the same recitation. There will be intermediate dates for group formation, choice of topic, choice of data and variables, preliminary results.

Participation is **strongly** encouraged. Attendance to the lectures and the recitations is mandatory. We will check the attendance at the beginning of each class. Students are allowed to skip **two lectures and two recitations** without any questions asked to accommodate any emergencies or important timing conflicts. However, presentation dates cannot be skipped and missing recitations that entail group work requires to coordinate with all group members.

Final exam will be cumulative, i.e. cover all the material of the course and will account for 30% of the final grade.

Because the course is graded based on a number of different assignments which puts less weight on any given assignment or exam, there will be no additional points given at the end of the course to "round up" the grade to the closest threshold.

If you work hard throughout, you will be able to achieve your target grade.

Online lectures

As long as lectures and recitations are held online, quizzes have to be taken, while having the camera turned on. Students are allowed to write solutions on a piece of paper and upload a picture of their solutions to Brightspace or alternatively use a Tablet.

Online lectures will not be recorded and uploaded. In case you have to miss a lecture, please try to follow the lecture using slides and the book (see schedule), inform yourself asking other students, and schedule an office hour to ask any questions regarding the material.

Laptops and notes

For in-person lectures, we ask you to not use laptops or tablets in class for two reasons. There is substantial evidence that learning outcomes are better without using laptops in classrooms. Moreover, we will use graphical and mathematical expressions on the board that are easier to copy using regular pen and paper. All materials will be uploaded before lectures.

Grades (summary):

<i>Activity</i>	<i>Grade Percentage</i>
<i>Quizzes</i>	50 %
<i>Final project</i>	20 %
<i>Final Exam</i>	30 %

Overall Score	Final Grade
100 – 94	A
90 – 93	A-
87 – 89	B+
84 – 86	B
80 – 83	B-
77 – 79	C+
74 – 76	C
70 – 73	C-
67 – 69	D+
65 – 66	D
Below 65	F

COURSE OVERVIEW

TOPIC 1: INTRODUCTION

Reading: Ch. 1, Moore, et al.

Part I: Introduction to statistics (Aug. 29)

- Course structure and goals
- What is statistics? And why should we care?

Part II: Viewing the data (one variable at a time) (Aug 29-31)

- Concepts of population, samples and random samples
- Types of variables: categorical and continuous variables
- Distribution I: graphic depictions
- Distribution II: numerical description
 - Mean, variance, percentiles
 - Population parameters vs sample statistics
 - Linear transformation
 - Normal distribution

TOPIC 2: RELATIONSHIPS IN THE DATA

Reading: Ch. 2, Moore, et al. Ch. 3 part on Experiments

Part I: Describing relationships (Sep. 5)

- Scatterplots
- Two-way tables

Part II: Correlations, regressions and causation (Sep. 5-19)

- Measuring relationships
 - Covariance and correlation
- Understanding regressions: linear model
- Interpreting regression's output
- Spurious correlation, correlation and causation
 - Omitted variable bias
 - Simpson's paradox
- Introducing multiple regressions (additional reading chapter from *Mastering Metrics*)
- Experiments
 - Research question
 - Design: treatment, controls, randomization, etc.
 - Pros and cons.
 - Best practices and ethics

Quiz 1 on Topics 1 and 2 during session on Sep. 26 to give you a weekend to refresh and study.

Deadline for project group formation and topic selection is on Sep. 22.

First STATA assignment due on September 22.

TOPIC 3: PROBABILITY

Reading: Ch. 4, Moore, et al.

Part I: Introduction to the probability theory (Sep. 21 and 28, on Sep 26 we have the first quiz)

- Basic concepts
 - Vocabulary
 - Uncertainty
 - Probability rules
 - Independence
- Computing probabilities:
 - Equally likely events
 - Conditional probability and compound events
 - Bayes' rule

Part II: Random variables (Sep. 28 Oct-3)

- Random variables
- Means and variances of random variables
- Law of large numbers

Part III: Probability distributions (Oct. 5-10)

- Interpreting probability distributions
- Bernoulli trials, Binomials and Poisson distributions

Quiz 2 on Topic 3 during the session on October 12.

Note: no classes on October 17-24.

TOPIC 4: STATISTICAL INFERENCE

Reading: Ch. 5, Moore, et al. for part I and Ch. 6-7 for part II.

Reading: Ch. 7, 10, Moore, et al. (for part III)

Part I: Introduction (Oct. 26)

- Introduction to inferential statistics
- Sampling variability
 - Bias and noise
- Sampling distributions
 - Mean
 - Counts
 - Proportions
- Central limit theorem

Part II: Core concepts (Oct 31 - Nov. 2-16)

- Confidence intervals and margins of error (Oct 31)
 - Hypothesis testing (Nov. 2-7)
 - Elements
 - Steps
 - Type I and II errors and power (Nov. 14-21)
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Part III: Introduction to inference for regressions (Nov. 28-Dec 7)

- T-student distribution (Nov. 28)
- Interpreting regressions' output (Nov. 30)
- T-tests (Dec. 5-7)

Quiz 3 on part I and II of Topic 4 during the session on Nov. 23

Quiz 4 on part III of Topic 4 during session on Dec. 12

Second STATA assignment due Nov. 3

Third STATA assignment due Nov. 24

Revision session on December 14.

TOPIC 5: DATA SOURCES

Reading: Ch. 3, Moore, et al.

- Data sources (Ad-hoc session covered during recitation)
- Sampling design:
 - Random, stratified and cluster sampling
 - Sampling weights and representative statistics
 - Issues: under-coverage, biases

Project presentations to be held during the recitations in the weeks of November 27- December 4.



Academic Integrity:

Compliance with and respect for academic integrity as defined by NYU and NYUAD policy is expected and enforced.

Note NYUAD's community commitment to integrity:

At NYU Abu Dhabi, a commitment to excellence, fairness, honesty, and respect within and outside the classroom is essential to maintaining the integrity of our community. By accepting membership in this community, students, faculty, and staff take responsibility for demonstrating these values in their own conduct and for recognizing and supporting these values in others. In turn, these values create a campus climate that encourages the free exchange of ideas, promotes scholarly excellence through active and creative thought, and allows community members to achieve and be recognized for achieving their highest potential. As part of the NYU global network, NYUAD students are also subject to NYU's all-school policy on [Academic Integrity for Students at NYU](#).

Alleged integrity violations are resolved using NYUAD's Academic Integrity Procedure:

<https://students.nyuad.nyu.edu/campus-life/student-policies/community-standards-policies/academic-integrity/>

Moses Center for Student Accessibility (CSA):

New York University is committed to providing equal educational opportunity and participation for students with disabilities. CSA works with NYU students to determine appropriate and reasonable accommodations that support equal access to a world-class education. Confidentiality is of the utmost importance. Disability-related information is never disclosed without student permission.

If you have any questions or would like to have further information about this, please visit the following link:

<https://www.nyu.edu/students/communities-and-groups/students-with-disabilities.html>

Contact:

mosescsa@nyu.edu

As a university student, you may experience a range of issues that can interfere with your ability to perform academically or impact your daily functioning, such as: heightened stress; anxiety; difficulty concentrating; sleep disturbance; strained relationships; grief and loss; personal struggles. If you have any well-being or mental health concerns, please visit the Counseling Center on the ground floor of the campus center from 9am-5pm Sunday - Thursday, or schedule an appointment to meet with a counselor by calling: 02-628-8100, or emailing: nyuad.healthcenter@nyu.edu. If you require mental health support outside of these hours call NYU's Wellness Exchange hotline at 02-628-5555, which is available 24 hours a day, 7 days a week. You can also utilize the Wellness Exchange mobile chat feature, details of which you can find on the student portal.
